

MATLAB HW2

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1 n^3 Complexity of Matrix Multiplication

The row-column formula is defined as follows:

$$(AB)_{j,k} = \sum_l a_{j,l} b_{l,k}$$

Note that we only care for $n \times n$ matrices in this example, so treat j and k as the same and equal to n .

Observe what occurs when computing the first row of AB . The first a is multiplied n times by a b term, as there are n rows of B . This is a complexity of n . This occurs n times as there are n columns in A . This is now a complexity of n^2 . Finally, this process must be repeated n times for each of the rows, which there are n of. Giving us our n^3 complexity.